PRINCE ABDULLAH BIN GHAZI FACULTY
OF COMMUNICATION AND INFORMATION TECHNOLOGY
Department of Computer Science

Course Name	:	Compiler Construction and programming languages
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Assignment [1]
Scanner for Python Language

Problem 1) Write a Lex specification file and use the Lex generator to construct a lexical analyzer for **Python** programming language with the following lexical properties:

1. NEWLINE, INDENT and DEDENT (unindent, it is a token that represents the fact that the current line is indented by less spaces than the one before)
2. Identifiers

```
IDENTIFIER::=  xid_start xid_continue*
```

Note) Assume that the xid_start is the letters of English language and xid_start is a digits and a letters of English language.

3. Keywords

```
False       await       else        import      pass
None        break       except      in          raise
True        class       finally     is          return
and         continue    for         lambda      try
as          def         from        nonlocal    while
assert      del         global      not         with
async       elif        if          or          yield
```

4. Literals:

```
LITERAL ::=  stringliteral | integer | floatnumber |
```

5. Operators:

The following tokens are operators:

```
+       -       *       **      /       //      %       @
<<      >>      &       |       ^       ~       :=
<       >       <=      >=      ==      !=
```

6. Delimiters

The following tokens serve as delimiters in the grammar:

```
(       )       [       ]       {       }
,       :       .       ;       @       =       ->
+=      -=      *=      /=      //=     %=      @=
```

```
&=          |=          ^=          >>=         <<=         **=
```

- **Problem 2)** write a Lex program to remove comments from Python program.
- **Problem 3)** write a Lex program to count the frequency of the given word in a file.
- **Problem 4)** write a Lex program to remove a stop words from text file.